

# Exact Coalescence and the ROF Semigroup Identity for Total-Variation Flow on a Finite Path

Route-A source-local certificate HCS-C279

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## Abstract

We solve the Euclidean total-variation gradient flow on every finite unweighted path. Maximal-monotone well-posedness, one-homogeneous dissipation, and a path Poincaré estimate give finite-time consensus at the initial mean.

## 1 Frozen path flow

Let  $P_n$  be the path with vertices  $1, \dots, n$ , where  $n \geq 1$ . On  $\mathbb{R}^n$  with its Euclidean inner product define

$$(Dx)_i = x_{i+1} - x_i, \quad J(x) = \|Dx\|_1 = \sum_{i=1}^{n-1} |x_{i+1} - x_i|. \quad (1)$$

We study the maximal-monotone clock

$$x'(t) \in -\partial J(x(t)), \quad x(0) = x^0. \quad (2)$$

This freezes unit edge weights and the ordinary vertex norm. It is neither an Euler time step nor a continuum limit. Maximal-monotone semigroups provide the general existence framework [1]; graph total-variation flow and its finite-time stabilization are treated in a broader setting by Mazón, Solera, and Toledo [4].

For  $a \in \mathbb{R}$ , let  $\text{Sgn}(a) = \{\text{sign } a\}$  if  $a \neq 0$  and  $\text{Sgn}(0) = [-1, 1]$ . The chain rule for (1) gives

$$\partial J(x) = \{D^\top z : z_i \in \text{Sgn}((Dx)_i)\}. \quad (3)$$

Since  $J$  is finite, continuous, and convex,  $\partial J$  is maximal monotone; there is a unique global absolutely continuous solution of (2).

Write  $\bar{x} = n^{-1} \sum_i x_i^0$  and  $\rho(t) = \|x(t) - \bar{x}\mathbf{1}\|_2$ . Every  $D^\top z$  is orthogonal to  $\mathbf{1}$ , so the mean is preserved. One-homogeneity implies  $\langle g, x \rangle = J(x)$  for  $g \in \partial J(x)$ . With  $g = -x'$  and the convex chain rule,

$$\frac{d}{dt} \frac{\rho^2}{2} = -J(x), \quad \frac{d}{dt} J(x) = -\|x'\|_2^2 \quad (4)$$

almost everywhere.

**Lemma 1** (Finite consensus). *The solution reaches  $\bar{x}\mathbf{1}$  by a time  $T$  satisfying*

$$T \leq \sqrt{n} \|x^0 - \bar{x}\mathbf{1}\|_2, \quad (5)$$

*and remains there.*

*Proof.* For each  $i$ , path telescoping and averaging give

$$|x_i - \bar{x}| \leq \frac{1}{n} \sum_j |x_i - x_j| \leq J(x).$$

Hence  $\rho \leq \sqrt{n} J(x)$ . While  $\rho > 0$ , the first identity in (4) yields  $\rho' = -J(x)/\rho \leq -1/\sqrt{n}$ . Integration proves (5). At a constant vector the minimal subgradient is zero, so uniqueness keeps the state constant.  $\square$

## References

- [1] H. Brezis, *Opérateurs maximaux monotones et semi-groupes de contractions dans les espaces de Hilbert*, North-Holland Mathematics Studies 5, 1973, MR0348562.
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